DO NOW — RECALL

Final specification and synoptic checkpoint

Name: _____ Class: _____ Date: _____

1. Name a PageRank property beyond counting inbound links.

2. What is a page fault?

3. What does an INNER JOIN return?

4. Search value 11 in [1,3,5,7,9,11,13], showing mid indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Weighted contributions from links, damping and iterative calculation.

2. Model answer

Required virtual memory page not resident in RAM.

3. Model answer

Rows matching the join condition.

4. Model answer

mid3 (7) then mid5 (11) found; index5.

By the end, you can…

• Use evidence rather than ticks alone

• Find links between papers

• Prioritise gaps

Vocabulary — know these terms

synoptic — Combining knowledge across topics.

retrieval — Reproduce knowledge from memory.

misconception — Incorrect conceptual understanding.

intervention — Specific teaching targeting a gap.

NEW LEARNING

Read, annotate and explain

Use evidence rather than ticks alone

A topic is secure when a learner can solve an unfamiliar example, not merely recognise its name.

Find links between papers

Paper 1 numerical representation and database knowledge can underpin Paper 2 programming and algorithm questions.

Prioritise gaps

Record the exact misconception or missing step so later Paper 1/2 reteaching sessions target observable weaknesses.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Evaluate the claim 'all sorts take O(n log n) in the worst case'.

Predict first

Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.

Check reasoning

Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State roles of PC, MAR, MDR, CIR and ACC, plus one bus role. [6]

Question 2

A sorted list [2,4,6,8,10,12,14] is searched for 12. Trace binary search, then explain worst-case Big O. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A school chooses AI scoring for pupil support. Give three contextual safeguards and explain risks. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Knowing a topic name alone proves that you can apply the associated algorithm.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.